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RESEARCH ARTICLE

A Single-Stage Photovoltaic Module Defect Detection Method Based on Optimized YOLOv8

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ABSTRACT Defect detection in photovoltaic (PV) modules presents significant challenges. In the presence of inconspicuous or small-scale defects, downsampling operations can cause features to vanish, and computational complexity increases as the model deepens, resulting in reduced detection accuracy, higher latency, and challenges in deploying the model on edge devices with limited computational resources. In response, this article proposes a single-stage object detection model based on YOLOv8, the PSA-PVdetector (PSA-det). The core innovation of PSA-det is the novel Partial Spatial Attention (PSA) mechanism, which integrates Partial Convolution (PConv) with Spatial Attention (SA) to optimize feature extraction and reduce computational overhead. For small-scale defect detection, a Multi-Channel Feature Fusion (MCFF) detection head is designed to enable finer-grained center prediction. Additionally, ShapeIoU is employed as the bounding box regression metric, enhancing the classical IoU by incorporating bounding box shape, thereby improving defect localization accuracy. Experimental results demonstrate that PSA-det achieves an mAP50 of 87.2% on the Panel-2 dataset and 72.0% on the Solar dataset. On the Nvidia Jetson Xavier NX DK platform, PSA-det achieves an inference latency as low as 2.6 ms, effectively balancing accuracy and real-time performance.

INDEX TERMS YOLOv8, photovoltaic defect detection, edge deployment, partial convolution, spatial attention.

I. INTRODUCTION

Solar energy is among the cleanest and most abundant renewable energy sources. Photovoltaic (PV) power generation technology, which utilizes the photovoltaic effect to convert sunlight directly into electricity, has rapidly developed worldwide and has become a critical component of energy structure transformation in many countries. The design, quality, and stability of PV modules directly impact their output efficiency, reliability, and long-term economic benefits [1], [2]. However, prolonged exposure of PV modules to harsh environmental conditions—such as wind, sunlight, temperature variations, humidity, and pollutant erosion—can easily result in various forms of damage and defects [3], including surface scratches, cell grid breaks,

glass panel contamination, and encapsulation material aging. These defects reduce the energy conversion efficiency of the module and, in severe cases, can also trigger safety hazards such as short circuits and hot spot effects [4]. Therefore, ensuring quality control prior to factory dispatch, as well as implementing robust operation and maintenance technologies post-deployment, is particularly crucial.

To ensure the quality of PV modules, electroluminescence (EL) inspection techniques are commonly employed to identify defects before the modules leave the factory [5]. However, EL inspection requires a specialized inspection environment, is expensive, and cannot be utilized post-installation. Once the modules are in use, RGB inspection techniques have become vital for surface defect detection due to their intuitive and efficient nature [6]. Traditional manual RGB inspection methods, though intuitive, are inefficient and heavily reliant on personal expertise [7]. In contrast,

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FIGURE 1. PV defect detection system.

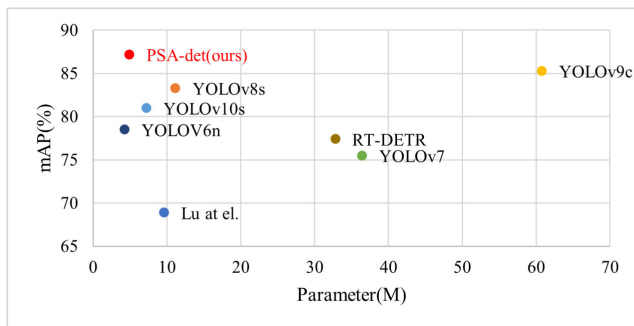


FIGURE 2. Comparison of previously advanced object detectors with the proposed method on the Panel-2 dataset.

UAV inspections, currently the most prevalent, are less affected by terrain and can operate under adverse conditions, thereby reducing downtime and maintenance costs [8], [9]. This paper focuses on RGB inspection techniques utilized in UAV inspections to identify surface defects that neither cause temperature changes nor affect the internal structure of the module but can reduce power generation efficiency. As illustrated in Fig. 1, a typical system includes a data sampling layer featuring a high-resolution camera-equipped UAV, an edge computing layer for automated defect detection through algorithms, and a cloud platform for monitoring and maintenance planning. Although UAV inspection technology significantly enhances inspection efficiency and coverage, it must address challenges such as real-time processing and data transmission delays to ensure prompt inspection feedback. Furthermore, to accurately identify and locate PV module defects, the system must rely on intelligent defect detection technology [10].

Recently, defect detection algorithms have significantly enhanced the ability to identify defects in PV modules [11]. Existing defect detection algorithms can be broadly categorized into two groups: classical computer vision-based methods and deep learning-based methods [12]. Classical methods typically rely on predefined templates to match features, which limits their effectiveness in handling complex and variable defects. Deep learning-based methods identify defects by automatically learning image features and are generally classified into two categories: two-stage algorithms, such as the RCNN (region-based) series [13], [14], [15]. Although two-stage algorithms offer high detection accuracy,

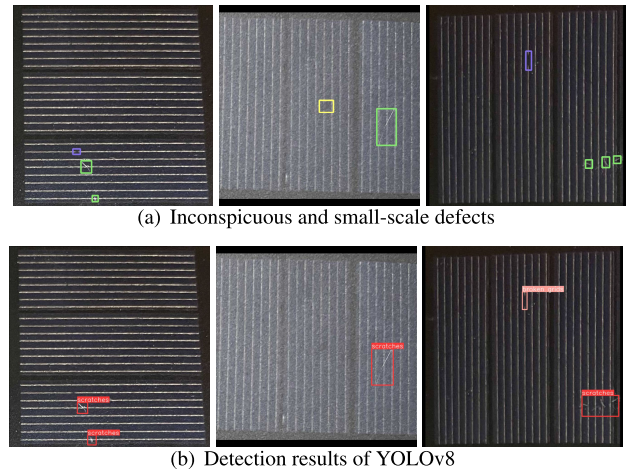


FIGURE 3. Three types of PV module defects are shown in (a), where the green box refers to scratches, purple refers to broken grids, and yellow refers to defaced. (b) shows the inference results of YOLOv8, with missed detections on images.

their detection speed is challenging to optimize due to the extensive computation required for extracting candidate regions. The second category is single-stage algorithms, such as the Single-Shot Multibox Detector (SSD) [16] and You Only Look Once (YOLO) series [17], [18], [19], [20], [21], [22], [23], [24]. The YOLO series provides several variants capable of achieving both high accuracy and rapid inspection speeds. Owing to its relatively simple architecture, ease of quantization and compression, and suitability for edge computing, the YOLO series has become a research focal point in industrial defect detection, aiming for real-time performance and high accuracy [25]. Notably, numerous EL defect detection efforts for PV modules have utilized the YOLO algorithm as a foundational approach [26]. As shown in Fig. 2 (with additional experimental results in Table 6), a performance comparison of related algorithms for RGB defect detection demonstrates that YOLOv8 significantly improves mean average precision (mAP) compared to previous enhanced algorithms. However, as the number of model network layers increases, the features of inconspicuous and small-scale PV module defects tend to vanish. As shown in Fig. 3, which illustrates three types of small-scale PV module defects, we observed certain limitations when using YOLOv8 for inference analysis. Specifically, the model fails to detect these types of defects. To investigate the reasons for this, we visualized the model's middle-layer output, as shown in Fig. 4. The visualized image reveals two core issues: First, there is feature map redundancy in the feature extraction stage, meaning that different feature maps contain a significant amount of duplicate information, which reduces the model's efficiency. Second, the model fails to effectively focus on critical defect-related features during feature fusion, leading to interference from redundant features and ultimately poor detection results. Therefore, retaining defective features during feature extraction is crucial.

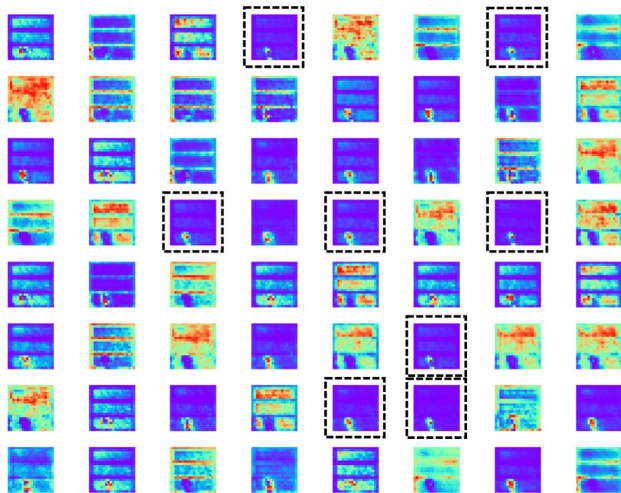


FIGURE 4. This feature map corresponds to the first map in Fig. 3, plot b. We have marked some of the similar feature maps with black boxes. In addition to the redundancy in the extracted features, it can be observed that the feature map focuses only on the features of the two defects and lacks the features of the broken grid defect.

Inspired by the need to improve feature extraction efficiency and enhance the model's focus on key features, this paper proposes an enhanced algorithm based on YOLOv8. The proposed enhancements target both the feature extraction process and the detection head. First, in the feature extraction phase, we introduce the C3-P and C2-SA modules, collectively termed "Partial Spatial Attention (PSA)," which combine Partial Convolution (PConv) [27] with the Spatial Attention (SA) mechanism [28]. The C3-P module utilizes PConv to selectively perform convolutional operations on specific channels of the input feature maps, leveraging similarity between feature maps to further reduce redundancy while retaining critical information from defective regions. The C2-SA module employs the SA mechanism to enable the model to focus on the most informative features, thereby reducing redundancy. Second, in the detection head section, a Multi-Channel Feature Fusion (MCFF) detection head is proposed to prevent feature disappearance for small-scale defects, with a small-scale detection head added for finer centroid prediction on a larger feature map. Finally, the PSA mechanism and MCFF are integrated into YOLOv8 to enhance the model's ability to detect inconspicuous and small-scale defects. The main contributions of this paper are as follows:

- 1) A PSA-PVdetector (PSA-det) based on YOLOv8 is proposed, incorporating both the PSA mechanism and an MCFF detection head. Additionally, the PSA-det engine model has been deployed on the Nvidia Jetson Xavier NX DK platform for inference. Experiments demonstrate that PSA-det achieves superior performance and reduced inference delay in detecting inconspicuous and small-scale defects.
- 2) The PSA mechanism is proposed, which combines the PConv and SA mechanisms to optimize feature

extraction, reduce redundant calculations, and make model inference more accurate and faster. Experiments show that the PSA mechanism is more efficient in feature extraction.

- 3) The MCFF detection head is proposed to address the inadequacies of standard detection heads in identifying small-scale defects. ShapeIoU is employed as the loss function for bounding box regression during training. Ablation experiments demonstrate that ShapeIoU converges more rapidly for bounding box regression compared to CIoU.

This article is organized as follows: Section II summarizes the related work. Section III gives the details of the proposed methods. Section IV describes the extensive experimental and ablation studies. Finally, Section V concludes this article.

II. RELATED WORK

While traditional manual inspection methods are time-consuming and reliant on experience, UAV technology combined with deep learning offers a novel approach to automated inspection. For example, Kuo et al. [29] proposed a method to automatically detect, classify, and locate defects in large-scale PV power plants using UAVs equipped with infrared and RGB cameras. This method employs a convolutional neural network (CNN) to identify and classify defects, such as hot spots, and then locates the PV modules using techniques such as color space conversion, morphological operations, and image segmentation. This method can automate the detection of various defect types; however, its generalization to complex environments remains a challenge. Cruz-Rojas et al. [30] proposed a novel comparative approach, evaluating three different methods to solve the photovoltaic (PV) module dust detection problem, primarily focusing on semantic segmentation techniques. These techniques can be used to automatically identify and assess the state of PV modules through image segmentation, reducing manual intervention; however, the effectiveness of this method for detecting other defects has yet to be examined. Li et al. [31] proposed an algorithmic solution for an unmanned inspection system for multi-defect detection in large-scale PV power plants. This solution first utilizes the Kirsch operator to identify abnormal regions in the image and then extracts defective features using a deep convolutional neural network (CNN) for each region. Finally, a Multi-Classification Support Vector Machine (MCSVM) is developed to integrate and classify these defective features. This method heavily relies on texture and boundaries, which may limit its effectiveness in detecting inconspicuous defects. Additionally, under strong lighting, high light spots caused by reflections on the PV module surface can easily lead to misrecognition.

In defect detection research, object detection techniques have become a focal research direction in this field due to their ability to accurately identify and localize specific defects within an image. For example, a BAF detector such

as that proposed by Su et al. based on the R-CNN model [32] achieved 88.07% mAP on the EL dataset PVEL-AD, but the high number of parameters in the model (120.87 M) limits its application in edge computing environments. Hong et al. proposed a novel framework [33] that combines visible and infrared images of photovoltaic (PV) modules for defect detection. The YOLOv5 algorithm is utilized for segmentation of PV modules, and then the segmented PV modules are classified using a deep residual network (ResNet). However, in some extreme cases, such as the loss of PV module image details due to strong reflections, it is difficult for this method to extract feature information and perform effective segmentation. Lu et al. proposed an optimized YOLOv5 model [26], which incorporates five data enhancement methods to expand the dataset and introduces the coordinate attention (CA) mechanism to replace the original YOLOv5 detection head with a decoupling head, resolving the conflict caused by classification and localization. However, two independent branches are used to train classification and localization, resulting in a lack of interaction between the two tasks. Sample assignment uses only IoU metrics, and only the best bounding box for localization is optimized during training, affecting classification accuracy during prediction.

Existing studies generally suffer from insensitivity to inconspicuous defect detection and high inference latency, especially when applied to edge computing devices. In view of this, this paper proposes the PSA-det model, which combines the PSA mechanism, the MCFF detection head, and the improved loss function design to improve the recognition accuracy and detection speed of unobtrusive defects while still maintaining good generalization performance across different batches of PV modules.

III. PROPOSED METHOD

In this section, we first introduce the architecture of the proposed PSA-det. Subsequently, we describe in detail the proposed PSA mechanism and the improvements made to the loss function.

A. PSA-det

The structure of PSA-det is shown in Fig. 5. The backbone network combines C2-SA and C3-P. C2-SA is used in the early stages to extract features and preserve their diversity, whereas C3-P is used in the later stages to screen for significant characteristics. The Spatial Pyramid Pooling-Fast (SPPF) structure [34] is utilized at the connection between the backbone and neck networks to capture picture features at multiple scales using MaxPooling kernels of various sizes, followed by feature splicing, and then C3-P to improve feature representation. After the features are multiscale fused, a decoupling head is utilized to train the classification and localization tasks independently.

In order to improve the accuracy of the detection of inconspicuous defects, we use a MCFF detection head. This detection head has higher-resolution output feature maps and contains feature extraction and prediction mechanisms

specific to small targets, using finer-grained center prediction and MCFF to ensure that the model maintains high sensitivity and localization accuracy when dealing with inconspicuous defects. In training, we use the TAL (task alignment learning) sample assignment approach [35], which considers higher-order combinations of classification score and IoU to measure the degree of task alignment when selecting positive and negative samples and describes the anchor's prediction quality in terms of the weighted product of classification score and IoU. Through training, it can direct the network to dynamically focus on high-quality positive samples. The TAL core formula is as follows:

$$t = s^\alpha \times u^\beta, \quad (1)$$

where s and u denote classification scores and IoU, respectively, α and β are hyperparameters (we chose the default parameter configuration: $\alpha = 0.5$, $\beta = 6.0$). For each GT box, we select 10 anchors with the maximum t as positive samples and select the rest of the anchors as negative samples.

B. PSA MECHANISM

In detecting small-scale photovoltaic (PV) module defects, image resolution diminishes as the network deepens, making it challenging to capture defect features. To address this issue and enhance feature extraction efficiency, we propose the PSA mechanism. This mechanism integrates the Cross Stage Partial (CSP) structure with partial convolution and spatial attention. Based on this mechanism, we introduce two modules: C3-P and C2-SA. The CSP structure, introduced in Cross Stage Partial Network (CSPNet) [36], processes input feature maps by splitting them into two distinct paths. Each path typically includes one or more "bottleneck" layers, and the resulting feature maps from both paths are recombined to leverage the complementary information obtained from each path. The C3-P and C2-SA modules adhere to the core principles of the CSP Bottleneck design to minimize computational redundancy while preserving model performance.

1) C3-P

When the current mainstream single-stage object detection model deals with inconspicuous defects, false detections and missed detections often occur. By visualizing the output feature maps of the middle layer of some models, as shown in Fig. 4, it is found that the YOLOv8 model has a lot of similar feature maps between different channels, especially in layers with a large number of output channels, which reduces the efficiency of feature extraction. In order to eliminate redundancy and improve the efficiency of feature extraction, this article proposes C3-P, which applies convolution operations to a specific subset of channels to reduce the amount of computation and improve the computational efficiency. Experiments show that the feature extraction efficiency of C3-P is significantly improved compared with C2-f, the feature extraction module of YOLOv8, and the visualization results are shown in Fig. 16.

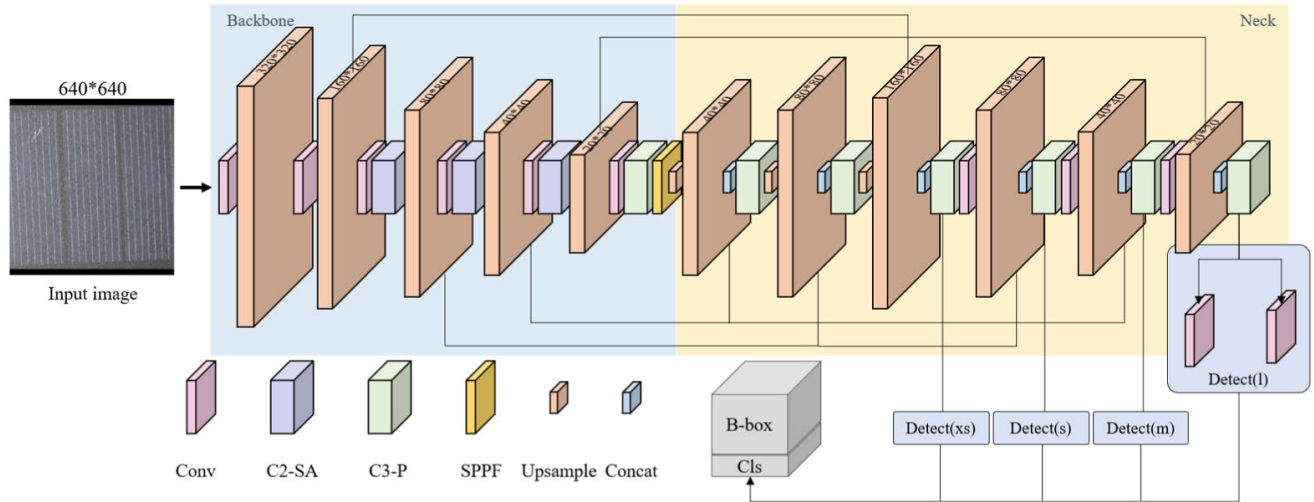


FIGURE 5. Structure of PSA-det.

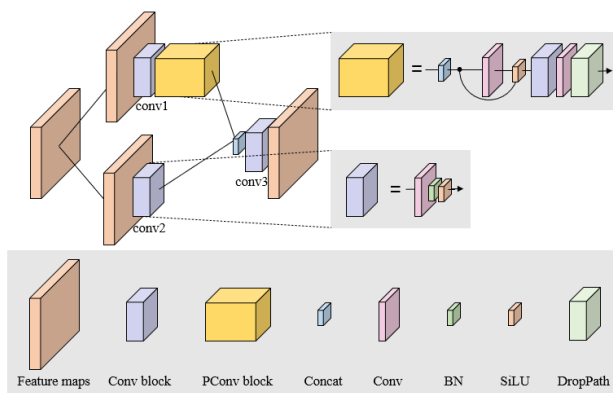


FIGURE 6. Structure of C3-P.

The structure of C3-P is shown in Fig. 6. First, C3-P receives the input feature maps, uses 1×1 convolution (cv1 and cv2) to adjust the number of channels of the input feature maps, respectively, and then divides the adjusted channels into two paths: one path enters the sequence consisting of the PConvblock, i.e., the multilayer perceptron module with PConv. The other path is processed directly through CV2 as a bypass branch to retain more original information. The features processed by the PConvblock sequence are spliced with the features of the bypass branch in the channel dimension to form a feature map with rich information. At this point, the spliced feature map is further processed by the last 1×1 convolution (cv3) to adjust back to the set number of channels.

Similarly, we visualize the feature maps in the middle layer of the model, as shown in Fig. 16, from which it can be seen that, compared with the original model, the extracted features are obviously diversified and the redundant information is

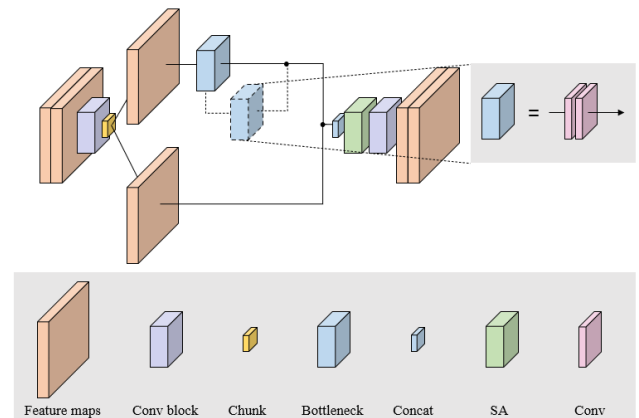


FIGURE 7. Structure of C2-SA.

reduced, which is conducive to the model learning more useful features. In addition, the C3-P module supports slicing in the inference stage, which avoids additional segmentation and splicing operations by directly processing part of the input through slicing and realizes efficient detection in an environment with limited computational resources.

2) C2-SA

For layers with fewer output channels, the C3-P structure would result in feature loss, so C2-SA is proposed, which is a combination of CSP and SA mechanism. The structure of C2-SA is shown in Fig. 7, which first transforms the feature map dimensions by means of a 1×1 convolution layer, and by doing so, it can obtain more feature representation space. The output is sliced into two parts of the same size, a design that allows feature information to flow in two paths and enhances feature diversity. Next, there is a series of bottleneck modules, each of which receives as input the results of the previous

processing. Each bottleneck contains two convolution layers, with possible jump connections (shortcuts) in between, for feature multiplexing and residual learning [37], enhancing the model's ability to learn features at different scales. The outputs of these modules are then spliced together with the original features, and the spliced features are applied with a SA module, which automatically adjusts the weights of the positions in the feature map according to the spatial context, allowing the model to adaptively focus on the most important parts of the image, suppressing extraneous background information, which can help the model better recognize the features under different illumination conditions. Finally, a 1×1 convolution layer controls the number of channels.

C. LOSS FUNCTION

The loss function of PSA-det consists of a classification loss function, a border regression loss function, and the formula is shown as follows:

$$Loss = L_{cls} + L_{box}. \quad (2)$$

The classification loss function combines sigmoid and BCE, whereas the bounding box regression function combines the ShapeIoU loss function and DFL [38]. The classic IoU loss function is described as follows:

$$IoU = \frac{|B \cap B^{gt}|}{|B \cup B^{gt}|}, \quad (3)$$

$$L_{IoU} = 1 - IoU, \quad (4)$$

where B denotes the prediction box and B^{gt} denotes the GT box. When the prediction box and GT box do not intersect, IoU is 1, L_{IoU} is 0, and the model cannot continue learning. To solve this problem, YOLOv8 uses CIOU to measure the degree of overlap, which adds a central normalized distance loss term and a shape loss term to IoU, taking into account the distance constraints between bounding boxes and the shape similarity between boxes, which is defined as follows:

$$CIOU = IoU - \frac{\rho^2(b, b^{gt})}{C^2} - \alpha v, \quad (5)$$

$$\alpha = \frac{v}{(1 - IOU) + v}, \quad (6)$$

$$v = \frac{4}{\pi^2} \left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h} \right)^2, \quad (7)$$

where w^{gt} , h^{gt} , w and h represent the GT box width, GT box height, anchor box width, and anchor box height, respectively. Although the CIOU loss function uses the bounding box aspect ratio as a penalty term in the loss function to keep the predicted box close to the real box over iterations, the aspect ratio metric is overly convoluted and slows down the regression convergence. To address this issue, this study introduces ShapeIoU [39], a bounding box regression approach that focuses on the impact of changes in the shape and size of the GT box on the bounding box

TABLE 1. Number of categories in the training set.

Classes	scratches	broken grids	defaced
Instances	712	704	684

regression using the following formula:

$$ww = \frac{2 \times (w^{gt})^{scale}}{(w^{gt})^{scale} + (h^{gt})^{scale}}, \quad (8)$$

$$hh = \frac{2 \times (h^{gt})^{scale}}{(w^{gt})^{scale} + (h^{gt})^{scale}}, \quad (9)$$

$$distance^{shape} = hh \left(\frac{x_c - x_c^{gt}}{c} \right)^2 + ww \left(\frac{y_c - y_c^{gt}}{c} \right)^2, \quad (10)$$

$$\Omega^{shape} = \sum_{t \in \{w, h\}} (1 - e^{-w_t})^\theta, \quad \theta = 4, \quad (11)$$

$$shape_{IoU} = IoU - distance^{shape} - 0.5 \times \Omega^{shape}, \quad (12)$$

$$w_w, w_h = hh \times \frac{|w - w^{gt}|}{\max(w, w^{gt})}, ww \times \frac{|h - h^{gt}|}{\max(h, h^{gt})}, \quad (13)$$

where $scale$ is the scaling factor ($scale = 0.5$ was finally chosen through comparative experiments, and the experimental results are shown in Table 5), which is related to the size of the target in the dataset, and ww and hh denote the weight coefficients in the horizontal and vertical directions, respectively, whose values are related to the shapes of the GT boxes. ShapeIoU eliminates the effects of the differences in the shapes and scales of the borders on the regression of the boxes by focusing on the shape scale attributes of the GT boxes themselves. Experiments show that ShapeIoU converges faster compared to CIOU.

IV. EXPERIMENTS AND RESULTS

A. DATASET

Two datasets are used in this article, each of which is described below.

1) PANEL-2

The Panel-2 dataset is available at <https://aistudio.baidu.com/datasetdetail/220416>, which contains many inconspicuous and small defects, a total of 2400 defect images, and is split into 1920 training sets, 240 validation sets, and 240 test sets in the ratio of 8:1:1. The dataset contains three types of defects: scratches, broken grids, and defaced, as shown in Fig. 8. The information is shown in Table 1.

2) SOLAR

The Solar dataset [40], available at <https://universe.roboflow.com/solar-abbaw/solar-msfcn>, covers samples of PV modules from different manufacturers around the world, taking into account a wide range of actual operating conditions. The dataset contains four categories: bird droppings, cracks, clean, and dust, as shown in Fig. 9. The

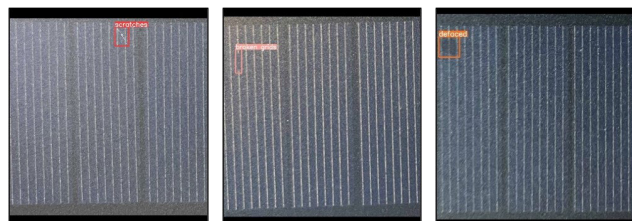


FIGURE 8. Visualization of Panel-2 dataset.

TABLE 2. Number of categories in the training set.

Classes	bird droppings	cracks	clean	dust
Instances	3351	2336	2213	2081

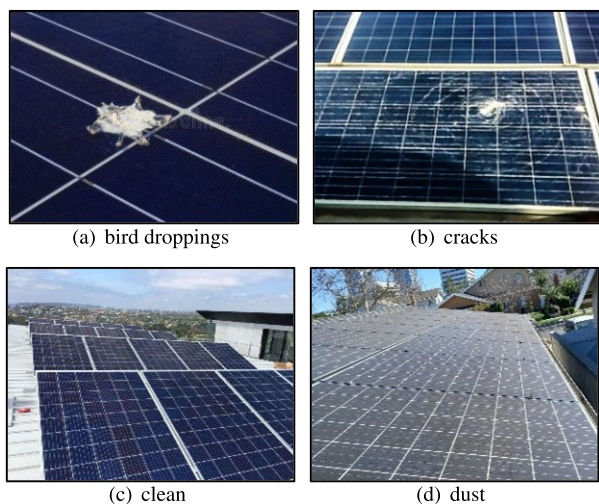


FIGURE 9. Visualization of Solar dataset.

dataset contains an instance segmentation data task and an object detection task. The dataset for the object detection task contains 5443 images, of which 4706 are in the training set, 467 in the validation set, and 270 in the test set. Table 2 shows the detailed data of Solar.

B. DATA ENHANCEMENT

The data enhancement techniques used in this article are described separately below.

1) RANDOM PERSPECTIVE

In order to simulate the PV module under tilted or non-vertical shooting viewpoints, we adopt a random perspective transformation, as shown in Fig. 10, where the main steps of the perspective transformation are to construct the perspective matrix and distort the image to realize the perspective effect. The hyperparameter *perspective*, which affects the degree of perspective distortion, is chosen to be 0.001, and as the value increases, the perspective effect will be more significant. A value of 0.001 can simulate the change

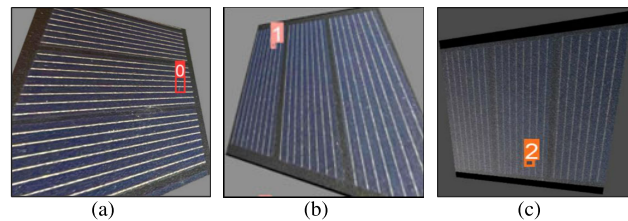


FIGURE 10. Perspective data enhancement method.

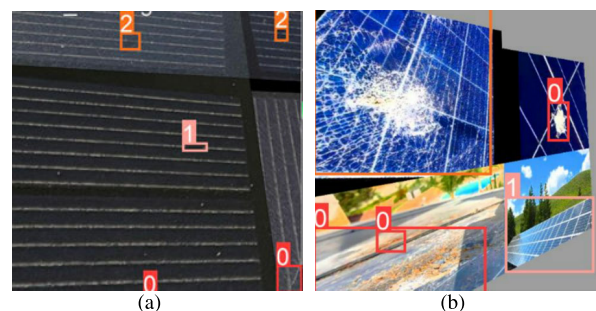


FIGURE 11. Mosaic data enhancement method.

TABLE 3. Configuration parameters.

Device	Configuration
System	Linux 5.15.0
CPU	Intel Core(TM) i5-1155G7 2.50GHz
GPU	GeForce RTX 4060 Ti 16GB
GPU accelerator	cuda12.0 cu113
Frames	PyTorch
IDE	VS Code
Python version	3.8.5

of perspective in the real world very well without producing too much distortion.

2) MOSAIC

Mosaic transform is also used for the training data, as shown in Fig. 11. Mosaic transform splices multiple images together to form a larger image through random cropping, so that the model learns more features in a batch, thus increasing the data diversity during model training and accelerating model convergence. In the last 10 epochs of training, we turn off mosaic because, toward the end of the training, the model gradually converges to a better performance state. Mosaic allows the model to be fine-tuned closer to the original data distribution, focusing on more details or hard samples.

C. EXPERIMENTAL ENVIRONMENT AND HYPERPARAMETER CONFIGURATION

The configuration parameters of the hardware and software platform for the algorithm implemented in this article are shown in Table 3.

The model is trained with initial weights, and the optimizer uses SGD with a learning rate of 0.01, momentum

TABLE 4. Detection performance at different resolutions.

Resolution	mAP50(%)	Latency(s)
480×480	85.8	5.28
544×544	86.4	5.38
640×640	87.2	5.75
736×736	85.3	7.19
800×800	84.1	7.34

TABLE 5. Impact of scale parameter on YOLOv8s model performance.

scale	mAP50(%)	mAP50-95(%)
0.0	83.8	43.8
0.5	84.6	43.8
1.0	84.0	43.4
1.2	83.2	42.9
1.5	82.4	43.5

of 0.937, batch size of 16, 150 epochs of training, and data augmentation is turned off in the last 10 epochs.

In order to ensure that the model can accurately detect small-scale defects on PV panels, the choice of image resolution is crucial. We explored the detection effect at different resolutions, as shown in Table 4. Experiments were conducted on 240 test images, and according to our results, the PSA-det model achieves its best performance (mAP50 = 87.2%) at a 640 × 640 resolution and is able to effectively identify most critical defects affecting performance.

In addition to the aforementioned hyperparameter configurations, this paper also addresses the critical hyperparameters of the loss function. By introducing parameters such as *scale*, ShapeIoU enhances the handling of bounding box shape and scale changes, thereby providing a more accurate assessment of the similarity between predicted boxes and ground truth boxes. We conducted comparative experiments on the YOLOv8s model using different values for the scale parameter. Our results indicate that the model performs optimally when *scale* = 0.5. The detailed experimental results are summarized in Table 5.

D. EVALUATION METRICS

In order to evaluate the performance of the model, this article uses precision (P), recall (R), the size of the model weights file (weights), the inference latency of TensorRT FP16 (latency), and mean average precision (mAP) as evaluation metrics. Precision describes the proportion of positive cases predicted by the model that are actually positive cases. Recall measures the proportion of all positive cases predicted by the model to all actually positive cases. The size of the model weight file is directly related to the complexity of the model. Models with a large number of parameters tend to require more computational resources and longer training times, as well as potentially leading to slower inference in the deployment phase. Inference latency measures the speed of the model in processing the images, i.e., the inference time required for each image. In this article,

we use the latency average of ten images as the inference latency. AP (average precision) is an important metric of the performance of each category's detection. The area under the precision-recall curve is calculated at different confidence thresholds, which is the AP. It integrates the precision and recall of the detection results; the higher the AP, the better the detection performance of the model in each category. mAP is the average of the AP of all categories, which is used to evaluate the comprehensive performance of the model on multi-category detection tasks. It is one of the most commonly used evaluation metrics in the field of object detection and reflects the overall detection accuracy of the model. The formula for the evaluation metric is as follows:

$$Precision = \frac{TP}{TP + FP}, \quad (14)$$

$$Recall = \frac{TP}{TP + FN}, \quad (15)$$

$$AP = \int_0^1 p(r)dr, \quad (16)$$

$$mAP = \frac{1}{n} \sum_{i=1}^n AP_i, \quad (17)$$

where *TP*, *FP*, and *FN* are the number of correct detections, the number of false detections, and the number of missed detections; *p(r)* denotes the P-R curve; and *n* is the category of defects.

E. RESULTS AND ANALYSIS

On the Panel-2 dataset, the comparison of training loss and validation loss of YOLOv8s and PSA-det is shown in Fig. 12. The loss of the PSA-det model is lower than that of YOLOv8, both in the training and validation phases. More critically, the model achieves a detection performance of 87.2% for mAP, as shown in Fig. 13.

F. COMPARISON OF MAINSTREAM OBJECT DETECTION MODELS

In order to verify the performance of the PSA-det model in the detection of insignificant and small defects, we have selected the mainstream object detection model and PSA-det for comparison experiments on the Panel-2 dataset, as shown in Table 6 and Fig. 14.

As shown in Table 7, we selected the following mainstream object detection models and conducted comparative experiments on the Solar dataset to verify the generalization ability of PSA-det in more complex scenarios.

G. COMPARISON OF DIFFERENT FEATURE EXTRACTION METHODS

In this subsection, we provide a comprehensive comparison between the proposed PSA mechanism and other feature extraction methods, focusing on two key aspects: accuracy and computational efficiency. To ensure a fair comparison, all experiments were conducted under the same hardware configuration and software environment using 240 test

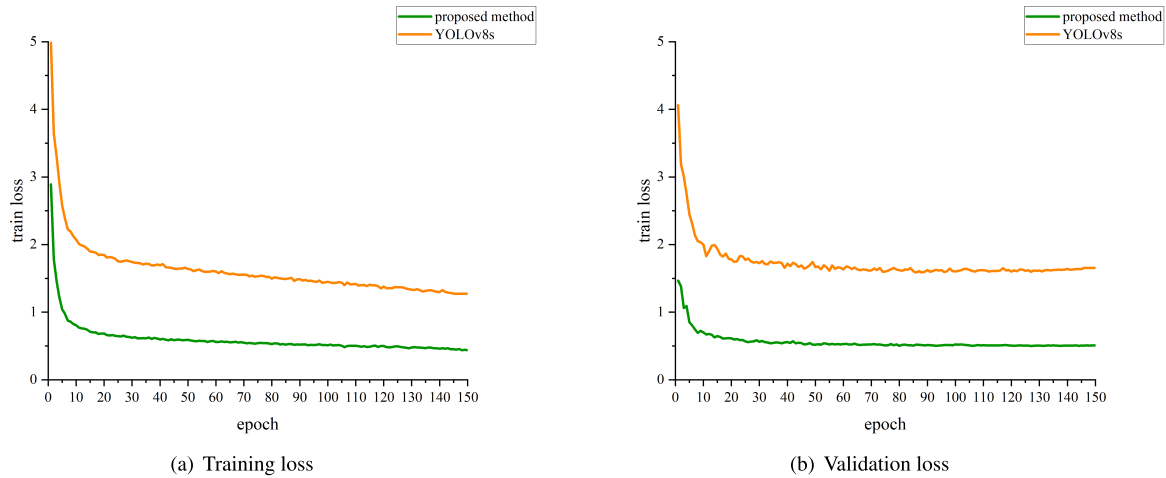


FIGURE 12. Comparison of training loss and validation loss.

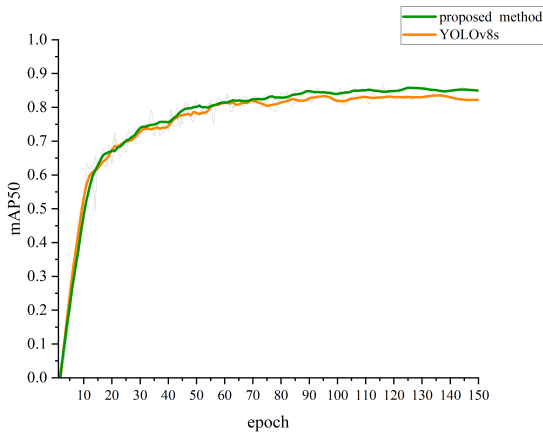


FIGURE 13. Comparison of mAP of YOLOv8s with the proposed model in the validation set.

TABLE 6. Comparison of mainstream object detection models.

Model	P	R	Weights (MB)	Latency (ms)	mAP50(%)
Lu et al.	69.4	68.0	19.5	-	68.9
YOLOv8s	82.5	81.6	21.4	3.9	83.3
YOLOv8m	90.6	77.4	49.6	7.4	85.8
YOLOv9c	82.6	84.1	116.0	9.4	85.3
YOLOv10s	78.6	78.0	16.5	-	81.0
YOLOv7	81.1	68.5	74.8	9.0	75.5
YOLOv6s	87.1	79.4	31.3	3.8	85.6
YOLOv6n	83.0	74.7	8.2	3.0	78.5
YOLOv5s	85.7	79.4	17.6	3.8	83.7
YOLOv3	80.1	78.4	23.2	16.8	79.6
RT-DETR	86.6	72.9	64.0	-	77.4
PSA-det	84.2	84.6	9.5	2.6	87.2

images from the Panel-2 dataset. The specific experimental results are shown in Table 8.

The PSA mechanism outperforms the other two feature extraction methods in terms of mAP50 while also demonstrating a significant advantage in inference latency.

TABLE 7. Comparison of mainstream object detection models.

Model	P	R	Weights (MB)	mAP50(%)
YOLOv8s	73.7	61.8	21.4	67.1
YOLOv8m	68.8	66.0	49.6	67.2
YOLOv9c	65.6	47.8	116.0	66.7
YOLOv10s	69.9	65.9	16.5	66.5
YOLOv6s	73.7	61.8	31.3	67.0
RT-DETR	74.8	64.4	64.0	61.0
PSA-det(s)	89.0	52.1	18.9	72.0

TABLE 8. Comparison of different feature extraction methods.

Feature Extraction Method	mAP50(%)	Latency(s)
PSA	87.2	5.75
Channel Attention	83.8	7.91
Multi-Scale Channel Attention	84.4	10.40

TABLE 9. Ablation experiment.

Models	PSA	ShapeIoU	MCFF	latency(ms)	mAP50(%)
YOLOv8s	×	×	×	3.9	83.3
YOLOv8s	✓	×	×	3.3	84.6
YOLOv8s	✓	✓	×	3.3	85.1
YOLOv8s	✓	✓	✓	2.6	87.2

H. ABLATION EXPERIMENTS

In order to further analyze the impact of different improvement methods on the performance of the YOLOv8s algorithm, three sets of experiments were designed on the Panel-2 dataset to analyze different improvement methods. In these experiments, we choose precision (P), recall (R), weight file size (weights), inference time (latency), and mAP50 of the model as the evaluation indexes of the experiments.

As shown in Table 9, compared with the first experiment and the second experiment, the number of model parameters is reduced by the PSA mechanism, which combines PConv

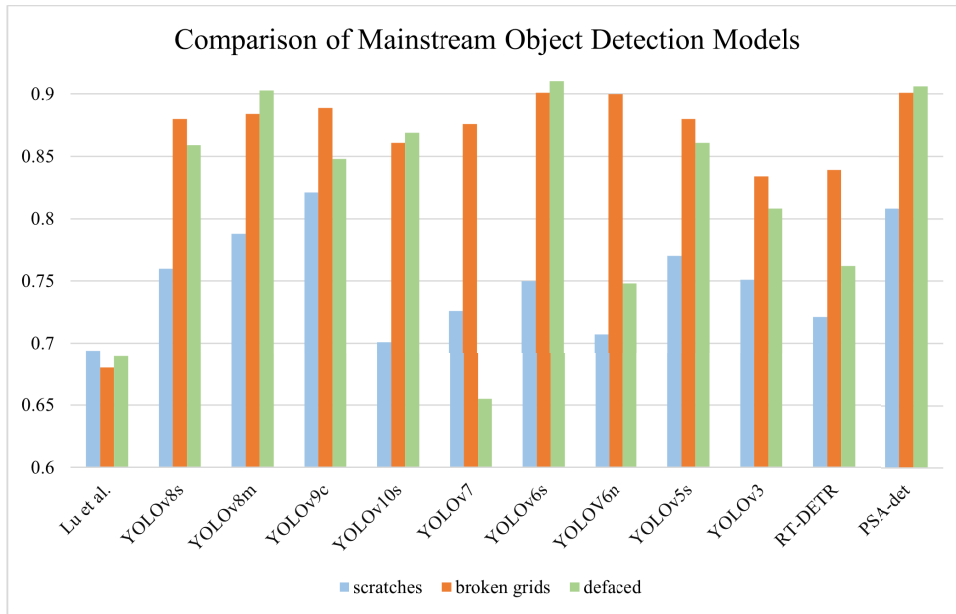


FIGURE 14. AP for each model.

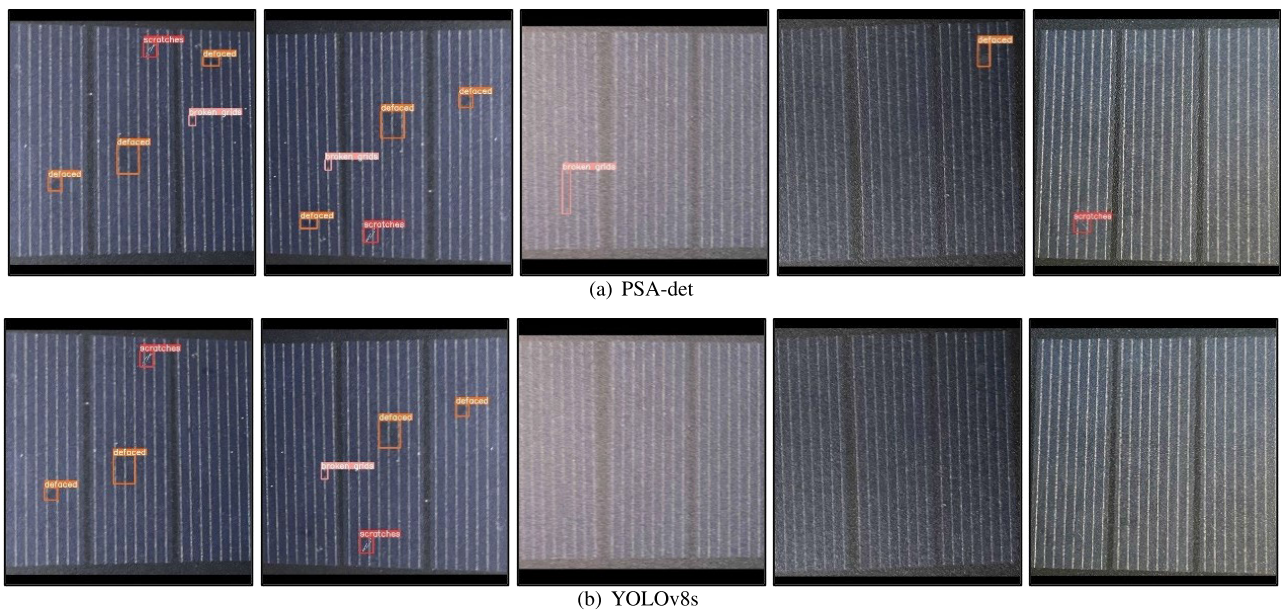


FIGURE 15. Visualization comparison of PSA-det and YOLOv8s.

and SA to reduce redundant feature representations while maintaining deep semantic information and model accuracy. The third experiment solves the problem of slow convergence in bounding box regression after introducing ShapeIoU. The fourth experiment introduces the MCFF mechanism, which selects a specific detection head to detect defects. This greatly reduces the size of the model weight file and improves the accuracy significantly. In addition, the number of parameters that can be learned during the training process is also greatly reduced, which makes the model converge faster.

I. VISUALIZATION

As shown in Fig. 15, we visualize some detection results of PSA-det and YOLOv8s. The first row shows the detection results of PSA-det, and the second row shows the detection results of YOLOv8s. In the first and second columns, YOLOv8s misses “defaced” and “broken grid”, while PSA-det can detect all defects. In the remaining three columns, YOLOv8s misses in all three categories, while PSA-det can detect all defects.

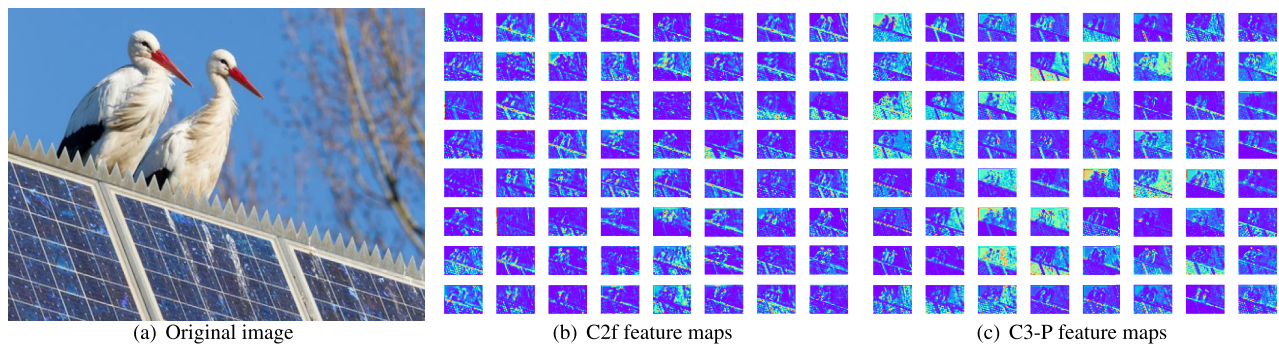


FIGURE 16. Visualization of model middle layer feature maps.

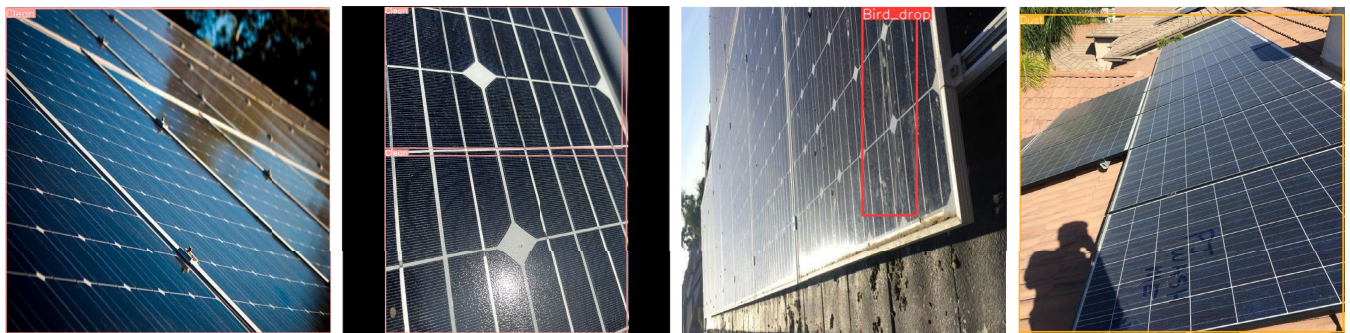


FIGURE 17. Visualization of PSA-det under glare and shadow conditions.

As shown in Fig. 16, we visualized the feature maps of the intermediate layers during model inference. The C3-P module was found to reduce the redundancy of feature maps compared to the C2f module, thereby demonstrating the effectiveness of the C3-P module.

In addition, to analyze PSA-det under different conditions, we selected samples with glare and shadows from the Solar dataset for visual analysis, as shown in Fig. 17. These results show that PSA-det accurately identifies various defects and does not misidentify reflective areas as defects. This highlights the model's strong performance in handling overexposed and shadowed areas.

V. CONCLUSION

In this paper, we introduce the PSA-det model, an enhanced variant of YOLOv8. The core innovation is the PSA mechanism, which integrates the Pconv and SA mechanisms to optimize feature extraction and minimize redundant computations. In the initial stage, the C2-SA module extracts features to preserve feature diversity. The C3-P module, employed in the mid-stage, retains crucial features and integrates both deep and shallow feature information in the later stage to enhance the backbone network's adaptability and feature extraction capabilities for photovoltaic module defects of varying sizes. Furthermore, to accurately detect small defects, an MCFF detection head is incorporated, preserving both semantic and global features of the up-sampled

feature map. The loss function design incorporates ShapeIoU to enhance the accuracy and convergence speed of bounding box regression. The PSA-det model's performance is evaluated through experiments on the Panel-2 and Solar datasets, demonstrated a mAP of 87.2% on the Panel-2 validation set, representing a 3.9% improvement over the YOLOv8s model. The mAP on the Solar validation set is 72.0%, reflecting a 4.9% improvement over the YOLOv8s model. Finally, the PSA-det's engine model was deployed to the Nvidia Jetson Xavier NX DK platform for inference, with a latency as low as 2.6 milliseconds per image, a significant reduction of 1.3 milliseconds compared to the YOLOv8s, which highlights its high performance in edge computing environments. The PSA-det model significantly reduces human resource expenditure and enhances the efficiency of automatic photovoltaic defect detection, making it suitable for widespread application in photovoltaic power plant inspections to ensure their efficient and safe operation.

Although the PSA-det proposed in this paper exhibits excellent detection performance on standard test sets, it still encounters several challenges in practical applications, particularly in complex environments and when handling highly intricate PV module defects. The following section provides a detailed analysis of the potential limitations of PSA-det and envisions future research directions to address these issues.

First, in complex backgrounds, PSA-det may be susceptible to interference, leading to missed defects. Complex backgrounds, characterized by rich textures, color variations,

and lighting effects, significantly increase the challenge for the model to differentiate defects from normal structures. Second, when handling highly complex PV component defects, PSA-det may face challenges related to insufficient feature fusion, especially when multiple types of defects coexist in the same module. These defects can share similar visual characteristics in images or overlap, complicating the model's task to distinguish and precisely locate each one. The concurrent presence of multiple defect types not only escalates the complexity of feature extraction but can also result in a loss of contextual information, thereby affecting sensitivity to specific defects. Third, the manually annotated data used in this study inevitably contains labeling noise. This noise arises from variations in subjective judgment and potential annotator errors, which can adversely affect model training, potentially leading to performance degradation or model deterioration.

For these limitations, although further expansion of the dataset is an intuitive solution, in practice, this approach is not always feasible due to the scarcity of defective samples and the high cost of labeling. Therefore, exploring small-sample learning techniques becomes a more realistic option [41], [42]. By introducing a meta-learning strategy, the model can quickly adapt to new types of defects with a small number of samples. Future research could combine small-sample learning and continuous learning techniques to reduce the dependence on large-scale labeled data while improving the generalization and accuracy of the model in PV defect detection tasks. In addition, building domain-specific large-scale models may be an effective way to address the problem of defect detection in PV components, where domain-specific large-scale models can utilize in-depth domain knowledge (e.g., materials, structures, and common defect types of PV components) to detect specific defects more accurately. Initial progress has been made in domain-specific large modeling techniques, especially in solving small-number learning problems [43]. Therefore, future work could explore the design of specialized models for PV defects, consider combining LLM and RAG techniques, and incorporate domain knowledge into the model construction and optimization process to improve model detection accuracy and generalization capability.

We are confident that through these efforts, PSA-det will be able to serve a wider spectrum of application scenarios. This advancement will provide the industry with more reliable and efficient solutions for detecting defects in PV modules.

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